Why

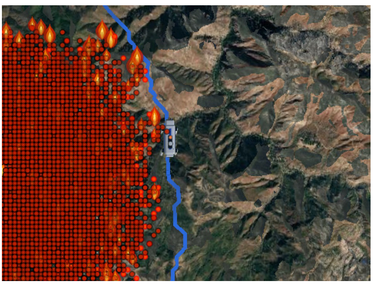
The edge of a wildfire is the most dangerous place for a firefighter.

Robot dogs (four-legged robots) can climb steep, rocky, smoky ground where trucks get stuck and drones can't see.

But a dog that steers by where the fire is now can walk into where it will be.

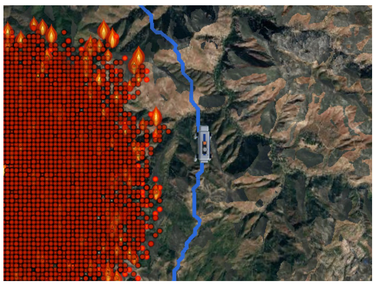
Plan on the predicted fire

Planned on the fire now



In the flames
An hour in, the fire has reached the route.

Planned on the prediction



Still clear
Same moment: this route keeps 100 m from the predicted fire.

The dog stays at least 100 m from where the fire is predicted to be, with 30 min in hand.

It scouts from the closest spot that keeps that distance and still has a way home. When its sensors show the real fire differs, the prediction reruns and the dog re-plans.

Simulator screenshots, Big Sur: same fire, same start and target, one hour after the dog sets out.

How we predict the fire

Fire history + conditions
246,000 NASA satellite fire detections from 30 wildfires; weather, terrain, land cover

→

Machine learning + fire physics
XGBoost adjusts a Rothermel spread model

→

When each spot burns
Every 50 m spot, in 15 min steps

Small and fast, so the dog can re-plan on the spot.

31 KB
The whole model. Smaller than one photo on this poster.

0.1 s
To predict all 360,000 spots on the map (desktop computer).

Anywhere
Small edge computers like a Raspberry Pi. No internet or GPU.

Close enough to help, safe enough to return

430 simulated missions per dog

	Reached a spot to scout	Caught by the fire
Reacts, keeps 500 m away	0%	0%
Reacts, keeps 200 m away	36%	0.9%
Predicts, no margin	100%	44%
Predicts, 100 m + 30 min + sensing	95%	0%

Reacting gets close only when it's lucky. Predicting knows when it's safe to go.

Each mission: walk from staging toward a target at the fire's edge, scout, come home. The predicting dog scouts from the closest spot that keeps 100 m from the predicted fire. Simulated on made-up terrain; the real fire runs hotter and drier than forecast, sometimes with a wind shift. No physical robot.